

Series TY-B — 5.6 K-factor Conventional (Old Style) Sprinklers Standard Response

IMPORTANT

Refer to Technical Data Sheet TFP2300 for warnings pertaining to regulatory and health information. Always refer to Technical Data Sheet TFP700 for the "INSTALLER WARNING" that provides cautions with respect to handling and installation of sprinkler systems and components. Improper handling and installation can permanently damage a sprinkler system or its components and cause the sprinkler to fail to operate in a fire situation or cause it to operate prematurely.

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docs.jci.com/tycofire/tpf662

General Description

The TYCO Series TY-B, 5.6 K-factor, Conventional Sprinklers (TY365) described in this data sheet are standard response -standard coverage, decorative 5 mm glass bulb type spray sprinklers. These sprinklers are intended to be installed either pendent or upright, and in either position, they produce a spherical water discharge pattern with approximately 50% of the discharge directed upwards and approximately 50% of the discharge directed downwards.

Conventional sprinklers are generally used with Ordinary and Extra High Hazard Class Systems, as defined by the automatic sprinkler system installation rules of the country and authority having jurisdiction. The NFPA permits the use of "Old Style Sprinklers" where special construction features require a unique water distribution; for the protection of fur vaults; or, the replacement of similar sprinklers that had been installed prior to 1955.

NOTICE

The TYCO Series TY-B 5.6 K-factor Conventional Sprinklers described herein must be installed and maintained in compliance with this document, as well as with the applicable standards of the NATIONAL FIRE PROTECTION ASSOCIATION, in addition to the standards of any other authorities having jurisdiction. Failure to do so may impair the performance of these devices.

The owner is responsible for maintaining their fire protection system and devices in proper operating condition. Contact the installing contractor or product manufacturer with any questions.

Sprinkler Identification Number (SIN)

TY365
TY3650 (LPCB Approved only)

Technical Data

Approvals
Refer to Table A

Maximum Working Pressure
175 psi (12,1 bar)

Discharge Coefficient
K = 5.6 GPM/psi^{1/2} (80,6 LPM/bar^{1/2})



Temperature Ratings

135°F (57°C)
155°F (68°C)
175°F (79°C)
200°F (93°C)
286°F (141°C)
360°F (182°C)

Finishes

Refer to Table B

Physical Characteristics

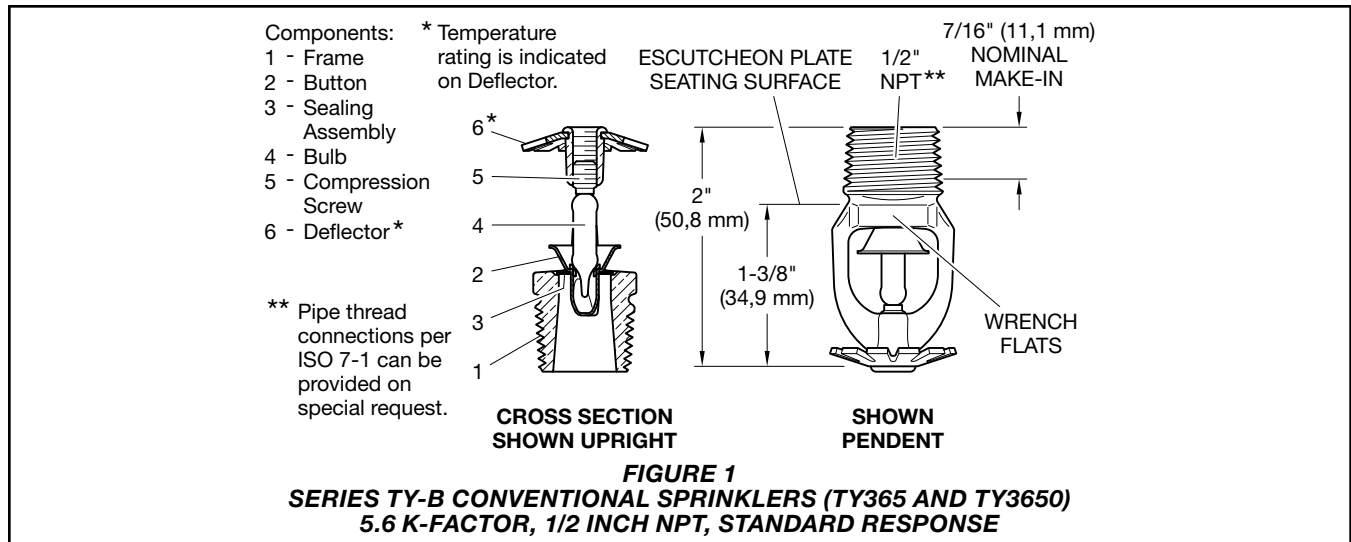
Frame	Bronze
Button	Brass/Copper
Sealing Assembly	Stainless Steel w/TEFLON
Bulb	Glass
Compression Screw	Bronze
Deflector	Bronze

Operation

The glass bulb contains a fluid which expands when exposed to heat. When the rated temperature is reached, the fluid expands sufficiently to shatter the glass bulb, allowing the sprinkler to activate and water to flow.

Design Criteria

The TYCO Series TY-B, 5.6 K-factor, Conventional Sprinklers (TY365) are intended for fire protection systems designed in accordance with the standard installation rules recognized by the applicable Listing or Approval agency (e.g., UL Listing is based on NFPA 13 requirements). The Series TY-B Conventional Sprinklers are only for non-recessed applications.



Installation

The TYCO Series TY-B, 5.6 K-factor, Conventional Sprinklers (TY365) must be installed in accordance with this section:

General Instructions

Do not install any bulb type sprinkler if the bulb is cracked or there is a loss of liquid from the bulb. With the sprinkler held horizontally, a small air bubble should be present. The diameter of the air bubble is approximately 1/16 in. (1,6 mm) for the 135°F (57°C) to 3/32 in. (2,4 mm) for the 360°F (182°C) temperature ratings.

A leak tight 1/2 in. NPT sprinkler joint should be obtained with a torque of 7 to 14 lb-ft (9,5 to 19,0 N-m). Higher levels of torque may distort the sprinkler inlet and cause leakage or impairment of the sprinkler.

Step 1. With pipe thread sealant applied to the pipe threads, hand tighten the sprinkler into the sprinkler fitting.

Step 2. Tighten the sprinkler into the sprinkler fitting using only the W-Type 6 Sprinkler Wrench (Figure 2), except that an 8 in. or 10 in. adjustable Crescent wrench is to be used for wax coated sprinklers. With reference to Figure 1, the W-Type 6 Sprinkler Wrench or the Crescent wrench, as applicable, is to be applied to the wrench flats.

Care and Maintenance

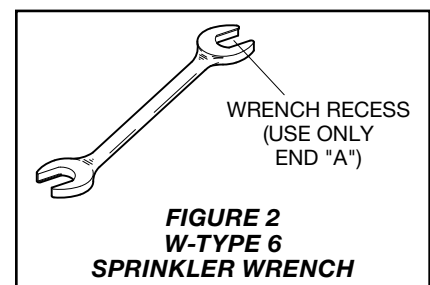
The TYCO Series TY-B, 5.6 K-factor, Conventional Sprinklers (TY365) must be maintained and serviced in accordance with this section.

Before closing a fire protection system main control valve for maintenance work on the fire protection system that it controls, obtain permission to shut down the affected fire protection systems from the proper authorities and notify all personnel who may be affected by this action.

Sprinklers which are found to be leaking or exhibiting visible signs of corrosion must be replaced.

Automatic sprinklers must never be painted, plated, coated, or otherwise altered after leaving the factory. Modified sprinklers must be replaced. Sprinklers that have been exposed to corrosive products of combustion, but have not operated, should be replaced if they cannot be completely cleaned by wiping the sprinkler with a cloth or by brushing it with a soft bristle brush.

Care must be exercised to avoid damage to the sprinklers before, during, and after installation. Sprinklers damaged by dropping, striking, wrench twist/slippage, or the like, must be replaced. Also, replace any sprinkler that has a cracked bulb or that has lost liquid from its bulb. Refer to the Installation section for additional information.



The owner is responsible for the inspection, testing, and maintenance of their fire protection system and devices in compliance with this document, as well as with the applicable standards of the NATIONAL FIRE PROTECTION ASSOCIATION (e.g., NFPA 25), in addition to the standards of any other authorities having jurisdiction. Contact the installing contractor or sprinkler manufacturer regarding any questions.

It is recommended that automatic sprinkler systems be inspected, tested, and maintained by a qualified Inspection Service in accordance with local requirements and/or national codes.

TYPE	TEMPERATURE RATING	BULB LIQUID COLOR	SPRINKLER FINISH (See Note 4)		
			NATURAL BRASS	CHROME PLATED	POLYESTER*
5.6K Conventional (TY365)	135°F (57°C)	Orange	1, 3, 4, 5		
	155°F (68°C)	Red			
	175°F (79°C)	Yellow			
	200°F (93°C)	Green			
	286°F (141°C)	Blue			
	360°F (182°C)	Mauve			
5.6K Conventional (TY3650)	135°F (57°C)	Orange	2		
	155°F (68°C)	Red			
	175°F (79°C)	Yellow			
	200°F (93°C)	Green			
	286°F (141°C)	Blue			
	360°F (182°C)	Mauve			

NOTES

1. Listed by Underwriters Laboratories, Inc. (UL)
2. Approved by the Loss Prevention Certification Board (LPCB Ref. No. 094a/05 & 007k/03)
3. VdS Approved (For details contact Johnson Controls, Enschede, Netherlands, Tel. 310530428-4444/Fax 31-53-428-3377)

4. Where Polyester Coated Sprinklers are noted to be UL Listed, the sprinklers are UL Listed as Corrosion Resistant Sprinklers only
5. EAC Approved
- * Frame and Deflector

TABLE A
LABORATORY LISTINGS AND APPROVALS

P/N* 77-XXX-X-XXX					
		SIN	SPRINKLER FINISH		TEMPERATURE RATINGS
573	5.6K CONVENTIONAL (1/2 in. NPT)	TY365	1	NATURAL BRASS	135 135°F (57°C)
573	5.6K CONVENTIONAL	TY3650**	3	PURE WHITE (RAL9010)*** POLYESTER	155 155°F (68°C)
			4	SIGNAL WHITE (RAL9003) POLYESTER	175 175°F (79°C)
			9	CHROME PLATED	200 200°F (93°C)
					286 286°F (141°C)
					360 360°F (182°C)

- * Use suffix "I" for ISO 7-1 connection; for example, 77-573-4-175-I
 ** LPCB approved Series TY-B Conventional Sprinkler assemblies only:
 Use suffix "L" for LPCB Approved assemblies (e.g., 77-573-4-175-L or 77-573-4-175-IL for LPCB Approved and ISO 7-1 connection)
 *** Eastern Hemisphere sales only

TABLE B
PART NUMBER SELECTION
SERIES TY-B CONVENTIONAL SPRINKLERS

Limited Warranty

For warranty terms and conditions, visit www.tyco-fire.com.

Ordering Procedure

Contact your local distributor for availability. When placing an order, indicate the full product name.

Sprinkler Assemblies with NPT Thread Connections

Specify: Series TY-B Conventional Sprinklers, SIN TY365, K=5.6, Standard Response, (specify) temperature rating, (specify) finish, P/N (specify, refer to Table B)

Sprinkler Wrench

Specify: W-Type 6 Sprinkler Wrench, P/N 56-000-6-387



TFP662 Change History Appendix

ISSUE DATE	NOTES
08-22	Page 1, updated QR code and URL; Page 4, changed corporate address and telephone number to 1467 Elmwood Avenue, Cranston, RI 02910 Telephone +1-401-781-8220, formerly 1400 Pennbrook Parkway, Lansdale, PA 19446 Telephone +1-215-362-0700.
04-22	Page 1, added QR code and URL to allow convenient access to electronic version from printed document; Page 3, Table A, TY365, added EAC Approved.
08-18	Updated Tyco® branding and document format; Added Johnson Controls copyright; Added disclaimer stating specifications and information subject to change without notice; Added reference to Regulatory and Health Warning Technical Data Sheet TFP2300.
08-16	Clarified approvals applicable to all temperature ratings offered: TY365 UL Listed, VdS Approved; TY3650 LPCB Approved.
03-16	Removed LPCB Approved from TY3650 360°F (182°C) Temperature Rating.
10-14	Added TY3650 with LPCB Approval only.
02-14	Changed Polyester finish sprinklers to indicate they are UL and C-UL Listed as corrosion resistant sprinklers.
06-13	Corrected Part Number Selection table, changing description to Conventional, formerly Pendent.
01-13	Updated note for special order ISO thread sprinklers, changing I code to suffix, formerly described as prefix; Corrected three-digit sprinkler type code to 573, formerly shown as 371.
12-12	New Technical Data Sheet TFP662 describes Series TY-B 5.6K Conventional Sprinkler.